Exploration and Analysis of Activation Functions

Bhavyesh Sajja Divyajyoti Panda Harshitha B R Hilay Mehta

Abstract

Activation functions are the cornerstone of artificial neural networks, introducing non-linearity and enabling the modelling of complex relationships. Various activations have been proposed in the literature, each coming with its advantages and drawbacks. In this project, we seek to analyze the performance of various state-of-the-art activation functions under five different datasets and two Deep Learning architectures, Extreme Learning Machine (ELM) and Multi-Layer Perceptron (MLP). Furthermore, we propose three novel activation functions. Results show that the proposed activation functions perform fairly well as compared to the state-of-the-art counterparts.

1. Introduction

Neural networks have emerged as a cornerstone in modern artificial intelligence, garnering immense popularity across various domains and expediting research in multiple fields, such as Computer Vision, Natural Language Processing, and Robotics. Their adaptability to diverse tasks coupled with their capability to continuously improve with more data has made them indispensable tools for tackling complex problems in science, industry, and beyond.

A key component of neural networks is the activation function used at each computation node. Activation functions introduce non-linearity in networks, enabling them to model complex functions. Multiple activation functions have been proposed in the literature which have been adopted for various tasks involving neural networks, such as the Sigmoid activation function, Rectified Linear Unit (ReLU), and Tanh-hyperbolic (Tanh). Each activation function lends different properties to the network. For instance, the Sigmoid activation function promotes interpretability by representing class scores as probability distributions for binary classification. The ReLU activation function provides fast inference and promotes generalization by implicit regularization (Vardi & Shamir, 2021). The multitude of choices for activation functions lends an element of versatility to neural networks. Each activation function also comes with its own set of caveats, such as the issues of dead neurons, vanishing/exploding gradients, etc.

The utility of activation functions is contingent upon the specific architecture they are used on, and the type of data used for training the network. In this project, the performance of various state-of-the-art activation functions has been analyzed across different neural network architectures and datasets, using multiple performance metrics. Furthermore, three novel activation functions have been proposed and compared against the state-of-the-art functions across various architectures and datasets.

2. Activation Function

In a neural network, activation functions are non-linear transformations used to transform the summed weighted input from a node into an output value to be fed to the next hidden layer or as output. A neural network can have multiple activation functions, with different activation functions at each layer.

As an example, for a single layer perceptron, let $\boldsymbol{X}$ indicate the input (including 1 for bias term) and $\boldsymbol{y}$ denote the output. Given activation function f, a weight vector $\boldsymbol{w}$ (including biases), and loss function L, training the perceptron aims to calculate the following:

$$\boldsymbol{w}^* = \arg\min_{\boldsymbol{w}} L(\boldsymbol{y}, f(\boldsymbol{X}\boldsymbol{w})) \tag{1}$$

2.1. Properties of Good Activation Functions

Properties of a good activation function include:

- **Non-linearity**: Activation function aims to produce non-linearity in the network, which models a non-linear response variable from features, allowing for complexities in the network.
- **Continuously differentiable**: A continuously differentiable function allows for calculating gradients for backpropagation in a neural network.
- **Range**: Range of activation functions provide a trade-off between stability and efficiency of training methods, where lower range leads to a more stable training with only limited weights being affected, while higher range leads to a more efficient training with more weights being affected.
- **Strict monotonicity**: A function f is defined to be strictly monotonous if $\forall x_1, x_2, x_3 \in \mathbb{R}, x_2, x_3 > x_1 \equiv (f(x_2) - f(x_1))(f(x_3) - f(x_1)) > 0$. A strictly monotonous activation function has two utilities:

- A strictly monotonous activation function has a non-zero gradient, which prevents stationarity in gradient updates, a problem known as dead neurons.
- A strictly monotonous activation function minimizes the amount of non-convexity in neural networks, which decreases the chance of getting stuck in local optima

- **Minimal vanishing/exploding gradients**: Vanishing and exploding gradients refer to the phenomenon when the gradients become too large or too small respectively. Both are undesirable, where the former leads to too little updates whereas the latter leads to too much and unstable updates.

2.2. State-of-the-art Activation Functions

To compare and benchmark the performance of novel activation functions, the following popular activation functions have been chosen. The respective plots are in Figure 1.

- **Sigmoid**(x) $= \frac{1}{1+e^{-x}}$
- **Tanh**(x) $= \frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}$
- **ReLU**(x) $= \max(0, x)$
- **Leaky-ReLU**(x) $= \max(0.01x, x)$
- **Step**(x) $= \mathbb{1}[x > 0]$

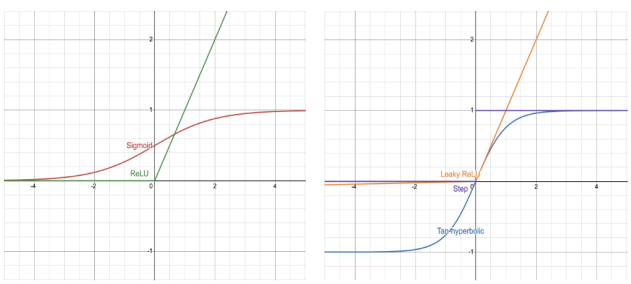

Figure 1. State-of-the-art activation functions

2.3. Novel Activation Functions

This section describes the proposed novel activations and the motivation behind their designs. For the experiment, n has been chosen as 0.5.

2.3.1. Roots

State-of-the-art activation functions each come with their own set of drawbacks. For example, the Sigmoid and Tanh Activation functions suffer from vanishing and exploding gradient problems, which inhibit proper training. Furthermore, the ReLU activation function is susceptible to generating dead neurons in the network that are never activated, hence bottlenecking the model's performance. To counteract the abovementioned problems, 'Roots' function (Panda et al., 2021) was proposed, defined as:

$$f(x) = |x|^n \operatorname{sign} x \tag{2}$$

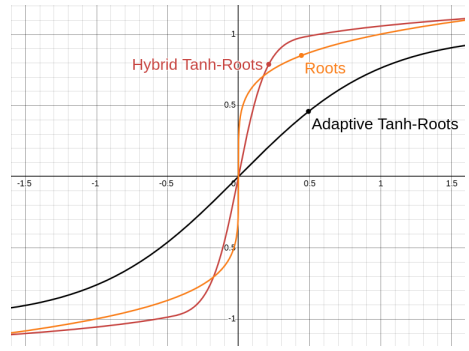

Figure 2. Novel Activation Functions ($n = 0.5$)

2.3.2. Hybrid Tanh-Roots

The function behaves as tanh within some neighbourhood of zero, and as roots function elsewhere. It is defined as:

$$f(x) = \begin{cases} \tanh x, & \text{if } |x| \leq \operatorname{sech}^{-1} n \\ (a|x|^n + b) \operatorname{sign} x, & \text{otherwise} \end{cases} \tag{3}$$

where

$$a = \frac{n}{(\operatorname{sech}^{-1} n)^{n-1}},\ b = \sqrt{1-n^2} - n \operatorname{sech}^{-1} n \tag{4}$$

2.3.3. Adaptive Tanh-Roots

The 'Adaptive Tanh-Roots' function goes one step further by utilizing $\tanh \frac{x}{n}$, providing flexibility in the steepness of the Tanh piece around zero. The factor n can be made adaptive, and backpropagation computation can be used to update n, but for simplicity, a fixed value of n has been used. The function is defined as:

$$f(x) = \begin{cases} \tanh \frac{x}{n}, & \text{if } |x| \leq n \operatorname{sech}^{-1} n \\ (a|x|^n + b) \operatorname{sign} x, & \text{otherwise} \end{cases} \tag{5}$$

where

$$a = \frac{1}{(n \operatorname{sech}^{-1} n)^{n-1}},\ b = \sqrt{1-n^2} - n \operatorname{sech}^{-1} n \tag{6}$$

2.4. Comparison of Activation Functions

Table 1 compares the activation functions across all the properties mentioned in section 2.1. All activation functions given are non-linear.

Gradients of activation functions have been plotted in Figure 3. The gradient of 'Roots' activation function does not decay as quickly as sigmoid and tanh activation functions, and therefore doesn't suffer from vanishing gradients problem as severely. However, the gradient of 'Roots' around 0 approaches infinity, which is undesirable. Therefore two more activation functions were developed which do not have either exploding or vanishing gradients problems.

Table 1. Comparison of activation functions across properties

ACTIVATION FUNCTION	CONTINUOUS	DIFFERENTIABLE	RANGE	STRICTLY MONOTONOUS
SIGMOID	✓	✓	$(0, 1)$	✓
TANH	✓	✓	$(-1, 1)$	✓
RELU	✓	✗	$[0, \infty)$	✗
LEAKY RELU	✓	✗	$(-\infty, \infty)$	✓
STEP	✗	✗	$\{0, 1\}$	✗
ROOTS	✓	✗	$(-\infty, \infty)$	✓
HYBRID TANH-ROOTS	✓	✓	$(-\infty, \infty)$	✓
ADAPTIVE TANH-ROOTS	✓	✓	$(-\infty, \infty)$	✓

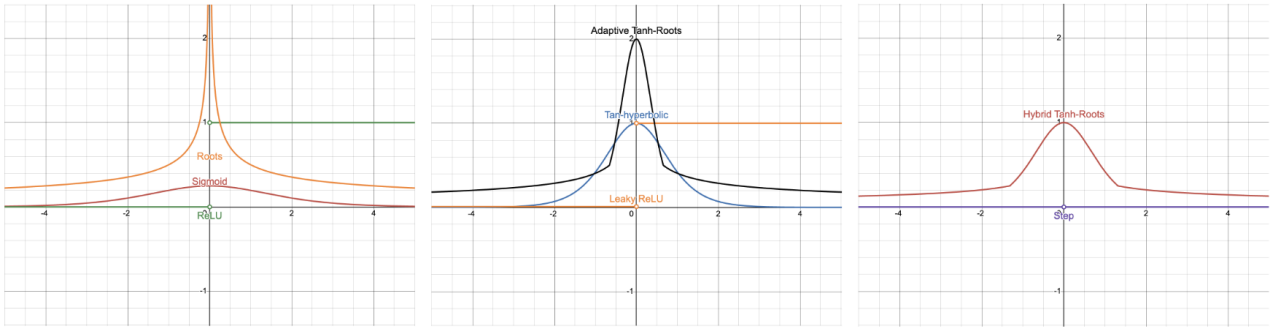

Figure 3. Gradient of Activation Functions ($n = 0.5$)

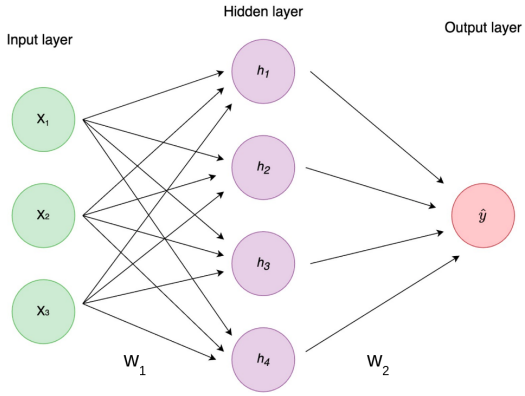

Figure 4. High-level architecture of Extreme Learning Machine and 1-hidden-layer Multilayer Perceptron

3. Neural Network Models

The experiments use Multi-Layer Perceptron (MLP) and Extreme Learning Machine (ELM) architectures, both with a single hidden layer, as illustrated in Figure 4. These are some of the simplest neural network architectures, and observations obtained on these models can be generalized and tested on more complex architectures as well.

3.1. Multi-Layer Perceptron(MLP)

A multi-layer perceptron (MLP) is a fundamental building block in artificial neural networks. It consists of an interconnected web of artificial neurons organized in at least three layers: an input layer, an output layer, and one or more hidden layers in between. The output of the network is $\hat{y} = W_2 \cdot Z$, where $Z = f(W_1 X)$ is the result of the hidden layer using an activation function f. The weights of an MLP are updated using backpropagation, with a suitable learning rate.

$$W_i^{(t+1)} \leftarrow W_i^{(t)} - \eta \frac{\partial L}{\partial W_i^{(t)}} \tag{7}$$

3.2. Extreme Learning Machine (ELM)

An Extreme Learning Machine (ELM) (Ding et al., 2014) is a Single-Hidden-Layer Feed-forward Neural network that requires no fine-tuning or backpropagation for training. ELMs are a realization of the Universal Approximation Theorem (Cybenko, 1989), i.e. a feedforward neural net with a single hidden layer can approximate any continuous function. A simple ELM comprises a single hidden layer, hence requiring two weight matrices. Consider the output of the ELM, $\hat{y} = W_2 \cdot Z$, where $Z = f(W_1 X)$ is the result of the hidden layer using an activation function f, and weight matrices between input and hidden layers, and hidden and output layers, are W_1 and W_2 respectively. To train an ELM, the weight matrix W_1 is randomly initialized. Hence, the optimization boils down to finding an assignment for W_2 to minimize the regularized l_2 loss, as shown in Equation 8, where y is the set of original labels. This is a regularized least-squares objective with a closed-form solution as shown in Equation 9.

$$W_2^* = \arg\min_{W_2} \lambda \|W_2\|_2^2 + \|y - \hat{y}\|_2^2 \tag{8}$$

$$W_2^* = ((Z^T Z) + \lambda I)^{-1} Z^T y \tag{9}$$

4. Workflow

Figure 5 illustrates the workflow for each dataset to run the experiments and generate results.

4.1. Datasets

For the experiment, five tabular Kaggle datasets with categorical datasets have been chosen, as described in Table 2. These datasets differ in terms of size, number of features, and number of classes. The variety in datasets allows us to compare activation functions under different settings. The CTG dataset contains two target features *NSP* and *CLASS*, with 3 and 10 target values respectively, out of which the former is considered as the target variable.

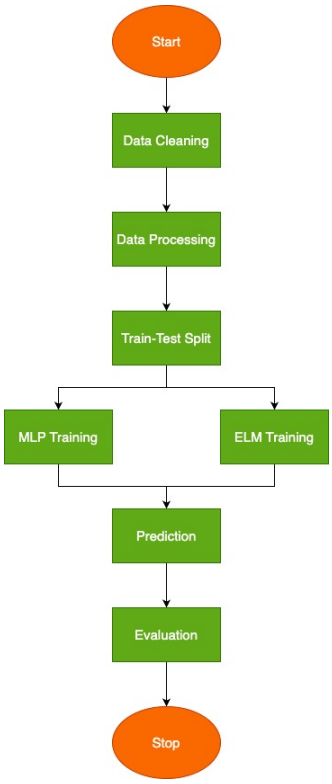

Figure 5. Workflow diagram

4.2. Data Cleaning and Processing

Data cleaning and processing steps included encoding categorical data into numerical values, standard scaling, handling missing and erroneous values, and removing features with near-zero variance. Feature selection was executed through two distinct methodologies: 1) Pearson's correlation coefficient to capture half of the features that are highly correlated with the target, and 2) Genetic Algorithm (GA) (Huang et al., 2007) to find optimal feature subset by iteratively evolving feature sets to minimize linear regression loss. These methods are representatives of filter based and wrapper based feature selection methods, and the choice may vary from use case to use case and dataset to dataset. The processed datasets were then partitioned into training and testing sets in an 80-20 split to prepare for model training.

4.3. Models

Each model was trained on the feature set obtained using the two feature selection techniques—Pearson's correlation coefficient and GA. To ensure the robustness and reproducibility of our results, each model was trained with five different seeds. For the ELM models, we varied the number of hidden units from 100 to 1500 to analyze performance implications. The MLP models were configured with a learning rate of 0.01. Both the models incorporate a L2 regularization factor of 0.01. Hidden units were varied from 25 to 250 and Cross-Entropy Loss was utilized. The batch size was set to 32. We also incorporated an early stopping mechanism with a patience parameter of 3 to prevent overfitting.

4.4. Performance Metrics

To evaluate the performance of activation functions, we considered the following metrics:

- **Training Microaccuracy**: Fraction of correct classifications on the training data after convergence.
- **Testing Microaccuracy**: Fraction of correct classifications on the testing data.
- **Training Time**: Time taken (in seconds) for training.
- **Epochs for Convergence (MLP only)**: Number of epochs needed for the MLP model to converge.

Micro-precision and micro-recall were also considered for the study, but they haven't been shown since they are numerically equal to micro-accuracy.

5. Results

In this section, we present the plots for the BRCA Dataset with Correlation-matrix-based feature selection. The observations and analyses are based on plots from all datasets in general. Other results and plots can be found in the Google Drive link at the end of the report.

5.1. ELM

The training accuracy of models increased as the number of hidden units was increased across all activation functions. This trend was consistent, demonstrating convergence in training accuracy irrespective of the activation function employed. However, the test accuracy portrayed a different behaviour; it generally stays constant, except in the case of CTG where a decrease in testing accuracy was seen, with an increase in the number of hidden units across all activation functions. These results suggest a potential overfitting issue

Table 2. Datasets used for experiments.

DATASET	SIZE (W/O NULLS)	#FEATURES	#CLASSES
DERMATOLOGY DATASET CLASSIFICATION (BOLAT, 2023)	358	35	6
BREAST CANCER WISCONSIN (BRCA) (WOLBERG & STREET, 1995)	569	30	2
OBESITY OR CVD RISK CLASSIFIER (ARAVIND, 2023)	2111	17	7
CARDIOTOCOGRAPHY (CTG) (CAMPOS & BERNARDES, 2010)	3126	21	3 OR 10
ARRHYTHMIA CLASSIFICATION DATASET (SAKIB, 2021)	712822	32	7

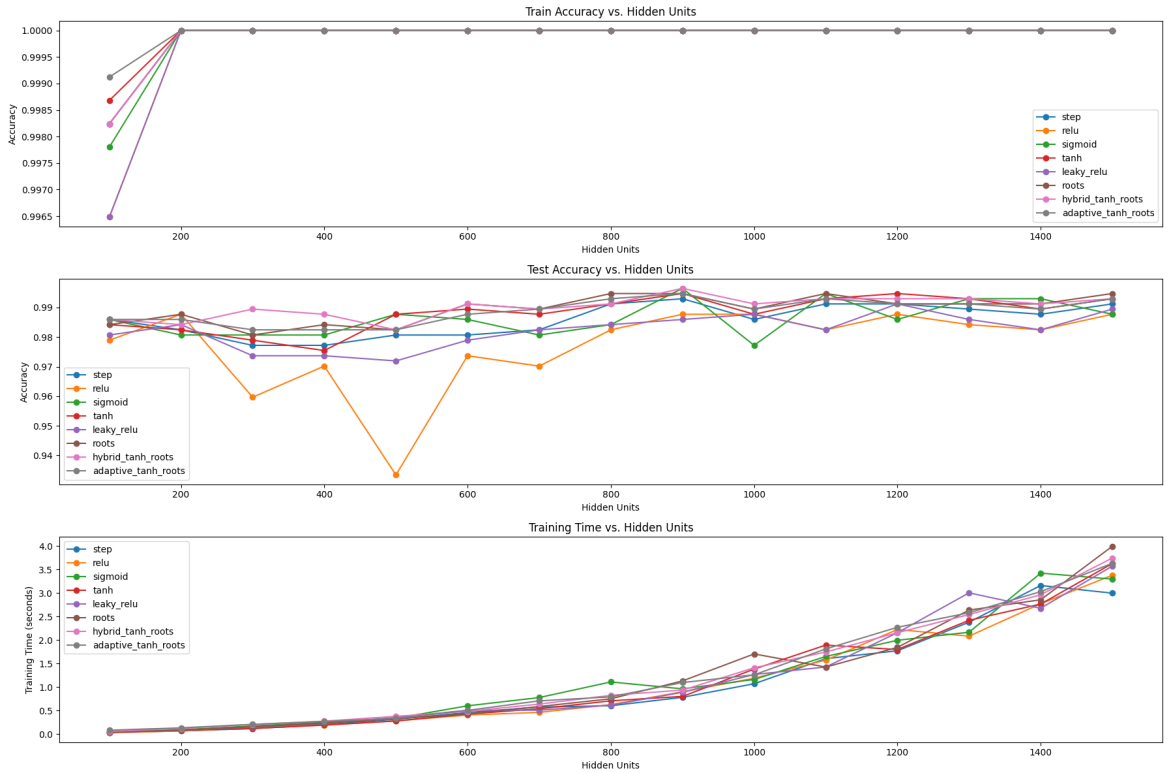

Figure 6. BRCA ELM Plot showing variation in metrics across varying hidden units. Figure best viewed zoomed in.

as the model complexity increased. Regarding generalization, the ReLU activation function showed comparatively poorer performance in most cases.

The training time was observed to generally increase with the number of hidden units, which is anticipated as larger models require more computations. There were minor fluctuations in this trend, which may be attributed to operational variations during the training process. The choice of activation function appeared to have minimal impact on the training time, suggesting that the complexity induced by larger models overshadows the computational differences between activation functions.

We also noticed that the Adaptive Tanh-Roots performs better than other activation functions on all datasets (for both Correlation and GA feature selection).

5.2. MLP

The investigation into the relationship between the number of hidden units and both training and testing accuracy revealed a generally low correlation. This suggests that the number of hidden units alone does not significantly influence the model's accuracy across most activation functions. However, notable exceptions were observed with Step functions, where an increasing trend in accuracy was identified, as the number of hidden units was increased. This trend underscores the sensitivity of these particular functions to model architecture changes.

Furthermore, the Sigmoid and Step functions both required more number of epochs for convergence, indicating these functions' potential inadequacies in simpler model architectures. In general, there is little correlation between training time and number of epochs till convergence with hidden units. Another general trend we observed was that the 'Hybrid Tanh-Roots' and 'Adaptive Tanh-Roots' functions performed reasonably well in comparison to the state-of-the-art activation functions. The original 'Roots' function does well only with CTG and BRCA datasets.

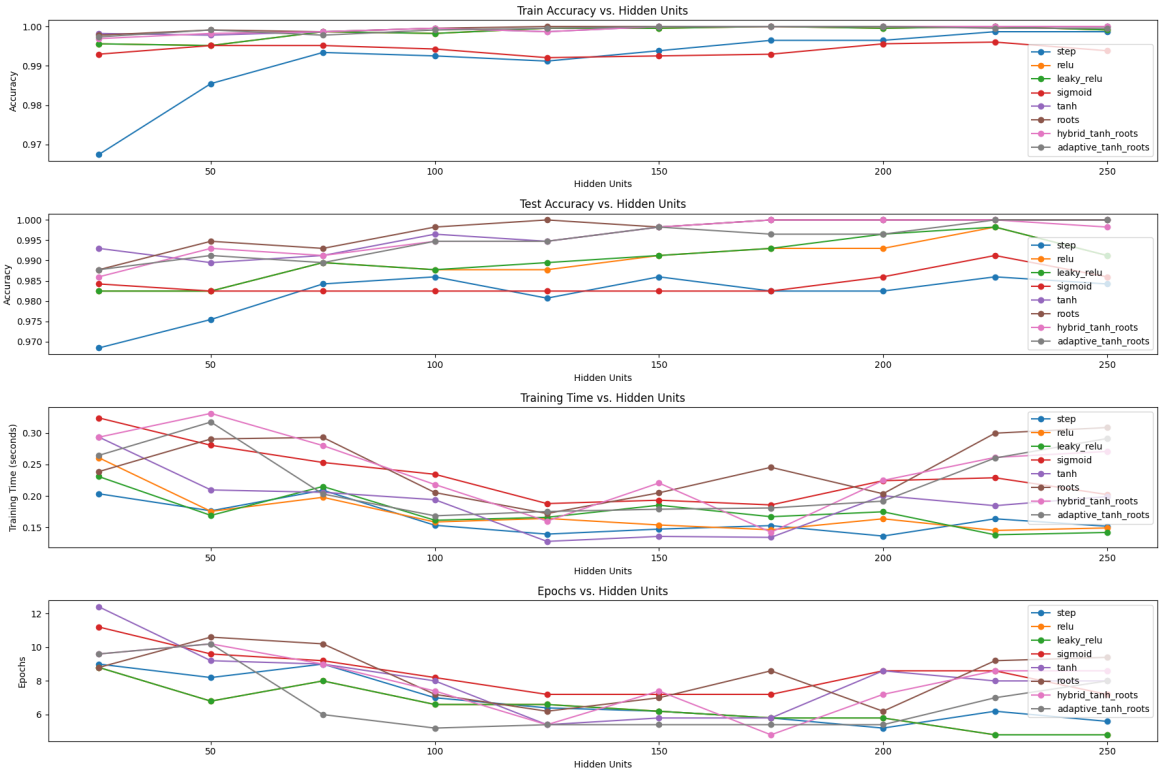

Figure 7. BRCA MLP Plot showing variation in metrics across varying hidden units. Figure best viewed zoomed in.

6. Challenges

The challenges encountered during the experiments are listed as below:

- **Arrhythmia Dataset**: During the project, significant challenges were encountered when attempting to perform the same analyses on the Arrhythmia dataset that were successfully applied to other datasets, despite utilizing advanced computational resources, including the CARC (Computational and Research Computing) facilities. This may be caused by its high dimensionality and the high number of data points, leading to exceptionally long processing times ($>$ 24 hours), particularly in Genetic Algorithms. This suggests that the analytical methods may need further optimization or that alternative, more efficient algorithms should be considered for processing such intricate datasets.

- **NaN problem**: The gradient of "Roots" blows up to infinity around 0 which leads to a severe exploding gradients problem in MLP, where updates are based on gradients. In such a case, loss is estimated to be extremely high, and hence the epoch is counted as not leading to a better result. Random small fluctuations are added to the weights to avoid getting this problem in the next epoch. ELM, not being based on gradients, does not suffer from this problem.

7. Contributions and Conclusion

In this project, we considered various activation functions to conduct a systematic and comprehensive analysis of their relative performances across varying architecture complexity and datasets. We proposed three novel activation functions to mitigate the issues posed by the state-of-the-art functions. Our experiments, which used various model architectures and datasets revealed that the proposed activation functions under certain conditions outperform the state-of-the-art activation functions. During our experiments, we faced challenges such as long processing times and the exploding gradient problem.
Future work could involve resolving these challenges, such as refining the proposed novel activation functions to improve time and memory requirements, implementing methods to monitor gradient values to make the functions more robust, and developing adaptive training methods that tune the activation function to better fit the training data.

8. Code and Dataset link

The code, datasets, and plots for all datasets and architectures can be found in the Google Drive link here.

References

Aravind. Obesity or cvd risk (classify/regressor/cluster), 2023. URL https://www.kaggle.com/datasets/aravindpcoder/obesity-or-cvd-risk-classifyregressorcluster.

Bolat, O. Dermatology dataset (multi-class classification), 2023. URL https://www.kaggle.com/datasets/olcaybolat1/dermatology-dataset-classification.

Campos, D. and Bernardes, J. Cardiotocography. UCI Machine Learning Repository, 2010. DOI: https://doi.org/10.24432/C51S4N.

Cybenko, G. Approximation by superpositions of a sigmoidal function. *Mathematics of control, signals and systems*, 2(4):303–314, 1989.

Ding, S., Xu, X., and Nie, R. Extreme learning machine and its applications. *Neural Computing and Applications*, 25:549–556, 2014.

Huang, J., Cai, Y., and Xu, X. A hybrid genetic algorithm for feature selection wrapper based on mutual information. *Pattern Recognition Letters*, 28(13):1825–1844, 2007. ISSN 0167-8655. doi: https://doi.org/10.1016/j.patrec.2007.05.011. URL https://www.sciencedirect.com/science/article/pii/S0167865507001754.

Panda, D., Panda, D., Dash, S. R., and Parida, S. Extreme learning machines with feature selection using ga for effective prediction of fetal heart disease: A novel approach. *Informatica*, 45(3), 2021.

Sakib, S. Ecg arrhythmia classification dataset, 2021. URL https://www.kaggle.com/datasets/sadmansakib7/ecg-arrhythmia-classification-dataset.

Vardi, G. and Shamir, O. Implicit regularization in relu networks with the square loss. In *Conference on Learning Theory*, pp. 4224–4258. PMLR, 2021.

Wolberg, William, M. O. S. N. and Street, W. Breast Cancer Wisconsin (Diagnostic). UCI Machine Learning Repository, 1995. DOI: https://doi.org/10.24432/C5DW2B.